經顱光生物調節（tPBMT）
用於犬認知功能障礙症候群（CCDS）

建議 Protocol、設定依據（Why）與作用原理

Transcranial Photobiomodulation Therapy for Canine Cognitive Dysfunction Syndrome

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結論

書上的建議只有一行：810 nm 近紅外線，手持件接觸頭頂（左右大腦半球都照），全程移動；皮膚表面輻照度 0.25–1.5 W/cm²、約 20 J/cm²；總能量 = 面積 × 20 J/cm²；第 1 週 2–3 次，再逐步減為每 2–3 週維持。

證據等級（請先看這裡）

目前沒有已發表的獸醫研究證實 tPBMT 對 CCDS 有效，理想參數也尚未確立。書中數字是從嚙齒類與人類阿茲海默症研究外推，再加上作者群的臨床經驗（「多數病例 4–6 週內症狀改善」是作者觀察，不是對照試驗）。建議向飼主說明這是外推、非標準的用法。

建議 Protocol 與設定依據

項目	建議設定	Why：為什麼這樣設
治療部位	頭頂（顱骨背側），左右大腦半球都照，手持件接觸皮膚	皮膚是光穿透的最大障礙；書中犬遺體研究顯示接觸式比離開皮膚到達深部的光更多
波長	810 nm（本院 GBOX 的 810 nm 通道；980 nm 設 0 W）	書中貓的示範用 810 nm；660／810／980／1064 nm 都有人研究，但各波長穿透力不同（受色素與顱骨上方肌肉厚度影響）
模式	CW（連續波）	書上以 CW 為基準；alpha（10 Hz）／gamma（40 Hz）脈衝有理論依據，但書中說好處尚待證實；若用 50% duty cycle，時間要加倍
輻照度（皮膚表面）	0.25–1.5 W/cm² = 功率 ÷ 光斑面積	目的是讓到達皮質的輻照度重現在人類身上安全有效的範圍；範圍上限同時避免皮膚過熱
劑量（皮膚表面）	約 20 J/cm²	光穿過頭皮、肌肉、顱骨會呈指數衰減，人類研究中頭皮 60 J/cm² 約只有 3.7% 到達硬腦膜；而皮質需要至少 5–10 J/cm²，所以表面要給得更多
總能量	治療面積 × 20 J/cm² 小型犬（約 20–25 cm²）約 400–500 J 中大型犬（50–70 cm²）1000–1400 J	書上中大型犬的數字；較小的病患依面積等比例縮小（小型犬面積為估計值，需實測）
操作	手持件持續移動；病患與操作者戴對應 810 nm 的護目鏡	四級雷射或高輻照度時，持續移動可避免單點過熱；810 nm 為不可見光，視網膜風險高
療程	第 1 週 2–3 次 → 每週 2 次 → 每週 1 次 → 每 2–3 週維持	CCDS 是慢性退化性疾病，先密集啟動、再逐步減量維持；書上另說每兩個月的維持治療有最好的持續效果
不適用	懷疑顱內持續出血、亞急性頭部外傷	PBMT 可能造成血流短暫增加，有顱內出血的風險

劑量怎麼算：用「水龍頭澆花」來想

功率是水龍頭開多大，能量是總共澆了多少水，光斑是手持件「印章」的大小，治療面積是整張「紙」，輻照度是印章下水噴得多猛，劑量是紙上每一小塊累積了多少水。

總能量（J）
= 治療面積 × 20 J/cm²
→
時間（s）
= 總能量 ÷ 功率
→
檢查輻照度
= 功率 ÷ 光斑面積
要在 0.25–1.5 W/cm²

兩個檢查

輻照度管安不安全（太高會燙），劑量管夠不夠（太低效果不足）。兩個檢查缺一不可：先用輻照度決定功率，再用劑量決定時間。

範例：不同體型怎麼設（GBOX，810 nm、CW）

體型	治療面積	總能量（20 J/cm²）	範例設定	時間
小型犬（約 4.5 kg）	約 22 cm²（估計，需實測）	約 440 J	1.5 W	約 293 s（約 5 分鐘）
中大型犬	50–70 cm²（書上）	1000–1400 J	4 W（光斑 ≥ 約 2.7 cm² 才不超過 1.5 W/cm²）	約 250–350 s

※ 時間 = 總能量 ÷ 功率。300 s 可平均分給左右兩側各約 150 s。小型犬面積為我的估計值；功率與時間都要等手持件光斑面積確認後才能定案。

GBOX 面板設定對照

面板欄位	設定	說明
810 nm POWER	依光斑面積決定（例：1.5 W）	功率範圍 = 光斑面積 × 0.25 到 × 1.5
980 nm POWER	0.0 W	不使用
LASER EMISSION MODES	CW	書上基準；脈衝（Repeat）的 Power 定義與單位需先看手冊
Time	總能量 ÷ 功率	例：440 J ÷ 1.5 W ≈ 293 s

定案前需要確認

① 手持件光斑面積：決定功率可用範圍（例：30 mm 鏡頭約 7.1 cm² → 約 1.8–10.6 W；10 mm 鏡頭約 0.79 cm² → 約 0.2–1.2 W；以上為鏡頭直徑換算，非原廠數字）。
② 本機規格（Device Settings 畫面）：Model Gbox、810 + 980 nm、最大 30 W、瞄準光 650 nm（< 5 mW）、脈衝 25 µs–6 s、1 Hz–20 kHz（可設 10／40 Hz）。功率上限不是限制，真正缺的是光斑面積。
③ 機器內建 "Canine Cognitive Dysfunction…" 預設內容（面板名稱被截斷）。
④ 建議：向飼主說明證據等級與同意事項，並在病歷記錄每次的功率、時間、總能量與反應。

認知功能障礙為什麼可能改善？

CCDS 是自然發生的退化性腦病，與人類阿茲海默症相似。書的邏輯是：血流受損、粒線體功能障礙、氧化壓力與發炎，正是這個疾病的核心，也剛好是光生物調節（PBMT）最擅長處理的環節。

① 腦裡發生了什麼

- 血管受損 → 腦血流不足
- 神經元粒線體功能障礙 → 能量（ATP）不足
- ß-類澱粉蛋白（Aß）沉積在血管與神經元周圍
- 發炎、氧化壓力、腦中甲醛（FA）升高
- 彼此加重，形成惡性循環

→ 樹突、突觸、神經元逐漸流失 → 認知退化

→

② PBMT 做了什麼

- 光被粒線體的細胞色素 c 氧化酶（CCO）吸收 → ATP 增加、粒線體功能恢復
- 抑制 ß-secretase、刺激降解 Aß 的酵素 → Aß 解聚（嚙齒類）
- 微膠細胞偏向抗發炎、神經保護的 M2 型
- 腦血管密度增加、發炎因子與 FA 下降（嚙齒類）

→

③ 預期結果

- 打斷「血流↓ → 粒線體↓ → Aß／發炎↑」的惡性循環
- 神經元的能量與環境改善
- 臨床上：定向力、睡眠週期、焦慮、夜間踱步等症狀改善

作者觀察：症狀常在前 1–2 次治療內開始緩解，多數病例 4–6 週內改善

請注意：這是「合理的推測」，不是已證實的療效

上面「PBMT 做了什麼」大多來自嚙齒類與機轉層級的研究，不是犬的臨床對照試驗。人類方面，書中提到近年有 tPBM 用於阿茲海默症、帕金森氏症、創傷性腦損傷等的研究，NEST 試驗（超過 1400 位患者）顯示 tPBM 安全，但療效結果不一致，原因是疾病不同與劑量不一致。所以：機轉合理、安全性資料良好，但犬 CCDS 的療效仍需要更多證據。

補充

CCDS 目前的處置（L-deprenyl、SAMe、抗氧化與粒線體輔因子飲食、必需脂肪酸、環境調整）可以和 tPBMT 並行；書中把 PBMT 定位為這些方法之外的選項，不是取代。

原文佐證（1）作用原理

以下為書中相關段落的截圖，附中文重點，方便核對。出處：Winkler & Miller (eds.), Laser Therapy in Veterinary Medicine: Photobiomodulation, 2nd ed., Wiley 2025, Ch.16。

dysfunction, inflammation, oxidative brain damage, and deposition of β-amyloid (Aβ) around blood vessels and neurons. Over time, these factors lead to progressive loss of dendrites, synapses, and neurons, followed by cognitive decline (Landsberg *et al.,* 2012; Chapagain *et al.,* 2018; Dewey *et al.,* 2019). As for AD in humans, there is no cure, although a variety of supplements, pharmacological agents, and environmental strategies have been used to treat CCDS, including L-deprenyl, S-Adenosyl methionine, diets fortified with antioxidants, mitochondrial cofactors, and essential fatty acids (Landsberg *et al.,* 2011; Manteca, 2011). While these methods may be helpful, authors and colleagues have observed that the use of transcranial PBMT seems to be very effective in improving the symptoms of CCDS within 4–6 weeks in most cases.

What is interesting from a PBMT standpoint is that compromised blood flow, mitochondrial dysfunction, oxidative stress, and inflammation are at the core of this disease process, and these all happen to be where PBMT shines as a treatment modality, having been extensively discussed in this and other chapters. Vascular compromise and inflammation result in the accumulation of neurotoxic Aβ, generating the production of ROS and proinflammatory cytokines (e.g., IL-1β, IL-6, and TNF-α), stimulating premature neuronal apoptosis via various signalling pathways, and increasing formaldehyde levels (FA) in the brain, all of which can lead to cellular and neuronal damage (De Taboada *et al.,* 2011; Da Luz Eltchechem *et al.,* 2017; Hamblin, 2019). While neuronal mitochondria are the primary source of energy production in the brain, when functioning normally they can also mitigate the toxic effects of Aβ on neuronal cellular function. Compromised cerebral blood flow and the toxic effects of Aβ contribute to mitochondrial dysfunction (Landsberg *et al.,* 2012; Chapagain *et al.,* 2018; Dewey *et al.,* 2019), which subsequently worsens the adverse effects of Aβ on cellular and mitochondrial function in a vicious cycle (Gonzalez-Lima *et al.,* 2014; Da Luz Eltchechem *et al.,* 2017; De La Torre, 2017; Lu *et al.,* 2017).

Studies have demonstrated the ability of PBMT to cause disaggregation of Aβ via suppression of β-secretase activity and stimulation of enzymes responsible for degrading Aβ peptides (De Taboada *et al.,* 2011; Da Luz Eltchechem *et al.,* 2017; Yue *et al.,* 2019; Zhang *et al.,* 2020; Tao *et al.,* 2021). Microglia play a central role in the generation and dissolution of Aβ in rodent AD models. PBMT has been shown to favor a shift between the two main types of activated microglia to the neuroprotective and anti-inflammatory M2 phenotype (Hamblin, 2019; Tao *et al.,* 2021). Rodent studies have demonstrated that tPBMT application decreased brain levels of Aβ, improved cognitive test scores, increased cerebral vascular density, and decreased FA (Huang *et al.,* 2020; Tao *et al.,* 2021).

作用原理（病理＋PBMT 機轉）

書 p.200 右欄

- CCDS 的病理是多因子的：血管受損、粒線體功能障礙、發炎、氧化壓力、Aβ 沉積。
- 作者與同儕觀察到 tPBMT 在多數病例 4–6 週內改善症狀（作者觀察）。
- 核心邏輯：血流、粒線體、氧化壓力、發炎，正是 PBMT 擅長處理的地方。
- 血管受損與發炎 → Aβ 堆積 → ROS 與促發炎細胞激素↑、神經元凋亡、甲醛↑，並與粒線體功能障礙互相加重（惡性循環）。
- PBMT 可使 Aβ 解聚、促使微膠細胞偏向 M2 型；嚙齒類研究見 Aβ 下降、認知測試改善、腦血管密度增加、甲醛下降。

原文佐證（2）證據等級與建議參數

While there are no veterinary studies published regarding the efficacy of use of tPBMT for the treatment of CCDS, one publication has discussed the pathophysiology and previously mentioned mechanisms by which PBMT may be beneficial (Dewey *et al.,* 2022). The ideal parameters for use of tPBMT for CCDS are not well established. However, data extrapolated from rodent and human clinical trials involving AD (Barrett and Gonzalez-Lima, 2013; Naeser *et al.,* 2014; Saltmarche *et al.,* 2017; Chan *et al.,* 2019; Salehpour *et al.,* 2019; Spera *et al.,* 2021) may be used as a basis of where to start. Though numerous wave-

證據等級

書 p.201 左欄上段

- 沒有已發表的獸醫研究探討 tPBMT 治療 CCDS 的療效。
- 理想參數尚未確立；可從嚙齒類與人類阿茲海默症試驗外推。
- 660、810、980、1064 nm 都被研究過，但穿透力不同，犬顱骨上方肌肉比嚙齒類與人類厚。

In a recent literature review examining PBM to prevent and treat AD, authors Enengl, Hamblin, and Dunkl (Enengl *et al.,* 2020) suggested that based on the available body of research on PBM to the brain (which took into consideration both safety and efficacy), application should achieve a dose in the minimum effective range between 5 and 10 J/cm^2 at the level of the cortex. Based on the exponential energy decrease during transcranial application, energy densities of 20–60 J/cm^2 applied to the human scalp would be suggested and should be somewhat similar in large to very large dogs. The same authors suggest that pulsing with alpha or gamma frequencies might add additional benefit to tPBM against AD, and a 50% duty cycle would necessitate a doubling of the calculated treatment time to achieve the same dose as with CW light.

The authors recommend that the handpiece be held on-contact with the dorsal surface of the skull, treating both the right and left hemispheres (Figures 16.4, 16.5), when treating CCDS. Laser safe eyewear for the patient is recommended and is discussed further in Chapters 4 and 5. When treating with a higher power laser (or at the higher end of the irradiance range listed below), the laser operator is encouraged to keep the treatment head in continuous motion at all times. A handheld class 3b laser may be held in place and used to treat point-to-point as an alternative. An irradiance between 0.250 and 1.5 W/cm^2 at the skin surface is recommended. The resultant irradiances at the cortical surface in dogs should replicate dosing that was safe and effective in humans (Ilic *et al.,* 2006) and tested in other species. Therefore, the laser operator must know the spot area of their laser handpiece as well as the power setting and duty cycle (if pulsed) for their laser to adequately calculate dose. In a medium- to large-size dog, treating over the dorsal calvarium, the size area may be between 50 and 70 cm^2 which would result in a total dose of 1000–1400 total joules delivered over the entire area if using

劑量、接觸方式、輻照度與總能量

書 p.201 右欄

- 皮質層最低有效劑量 5–10 J/cm²（Enengl 等人 2020，人類文獻回顧）。
- 經顱能量呈指數衰減，人類頭皮建議 20–60 J/cm²，大型到超大型犬可相似。
- alpha／gamma 脈衝可能有額外益處；50% duty cycle 時間要加倍。
- 手持件接觸顱骨背側，左右半球都治療；建議為病患配戴護目鏡。
- 高功率或輻照度在高端時，治療頭要全程持續移動；三級 B 雷射可逐點定位。
- 皮膚表面輻照度 0.250–1.5 W/cm²；必須知道光斑面積、功率與 duty cycle 才能計算劑量。
- 中大型犬頭頂 50–70 cm²，以 20 J/cm² 計算，總劑量約 1000–1400 J（下一頁接續）。

原文佐證（3）療程頻率與安全

a fluence of 20 J/cm^2 (scale down for smaller patients and treatment size areas accordingly). As with most chronic, degenerative conditions, PBMT should be started with 2–3 sessions the first week, if possible, weaning to a transition phase of less-frequent treatments (twice weekly, then once weekly), and continued once every 2–3 weeks as needed, keeping clinical signs at a minimum. Treatments are painless and well tolerated. Some colleagues have noted that the patient may seem a little tired for the first 24 hours, but resolution of clinical signs usually begins within the first one or two treatments and gets progressively better in the first month (Godine, 2017). Bimonthly maintenance treatments offer the optimal chance for continued efficacy Figure 16.6.

療程頻率與耐受性

書 p.202 左欄

- 接續上頁：以 20 J/cm² 計算，較小的病患依面積縮小。
- 慢性退化性疾病：第 1 週 2–3 次，之後減為每週 2 次、每週 1 次，再每 2–3 週維持，以症狀最小化為準。
- 治療無痛、耐受良好；部分病患前 24 小時可能稍微疲倦。
- 症狀通常在前 1–2 次治療內開始緩解，第一個月逐步進步。
- 每兩個月的維持治療最有機會持續維持療效。

Animal-model studies using rats have demonstrated that transcranial near-infrared laser application may provide benefit in cases of acute traumatic brain injuries (Xuan *et al.*, 2013). The same parameters as used for treating CCDS can be used to treat traumatic brain injury. The frequency of treatment should be every 12–24 hours as needed for aiding in resolution of clinical signs. PBMT can decrease intracranial inflammation, decrease oxidative stress, and help stabilize damaged membranes. It is not recommended that subacute head trauma cases be treated with PBMT due to the possibility of intracranial hemorrhage; PBMT may cause a transient increase in blood flow. If there is any reason to suspect that intracranial bleeding is ongoing, PBMT should not be used.

不要用的情況

書 p.202 右欄上段（創傷性腦損傷一節）

- 創傷性腦損傷可使用與 CCDS 相同的參數，頻率每 12–24 小時一次。
- 不建議亞急性頭部外傷使用：PBMT 可能造成血流短暫增加，有顱內出血的風險。
- 只要有理由懷疑顱內持續出血，就不應使用 PBMT。

資料來源與使用說明

Winkler CJ, Miller LA (eds.). Laser Therapy in Veterinary Medicine: Photobiomodulation. 2nd ed. Hoboken: John Wiley & Sons; 2025. Ch.16 Neurological Conditions, pp.198–203。書中引用的 CCDS 原始文獻包括 Dewey et al. 2022（Open Vet J 12:256–263）、Enengl et al. 2020、Schiffer et al. 2009、Piao et al. 2017，本簡報未逐篇核對原文。

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